

Iliya Mirzaei

Stony Brook, NY

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SUMMARY

Ph.D. student at Stony Brook who got hooked on a specific problem: how MEV reshapes transaction fairness in DAG-based BFT consensus. My research targets ordering vulnerabilities in protocols like Mysticeti and Bullshark, combining formal analysis with real implementation in Rust, published as three preprints and a peer-reviewed poster. Before the PhD, I spent 3+ years shipping production software to 200K+ users across multiple companies, and that engineering grounding keeps my research from staying theoretical. Looking for Summer 2027 roles in blockchain infrastructure, consensus protocols, or security.

EDUCATION

Ph.D. in Computer Science, *Stony Brook University* (GPA: 3.83/4.0) Aug 2024 - May 2029 (Expected)

- Advisor: [Prof. Mohammad Amiri](#). Research focus: Distributed Systems & Consensus Protocols
- **Relevant Coursework:** Distributed Systems, Database Systems, Computer Networks, Data Visualization, Data Science

B.Sc. in Computer Engineering, (*IUST*) (GPA: 3.79/4.0) 2019 - Feb 2024

- Top 10% of class, Top 4 university in country. Senior Thesis: Fraud Detection SDK (20/20).

INTERESTS

Distributed Systems, Consensus Protocols, Blockchains, and Software Engineering

PUBLICATIONS

- **Fair on the Surface: Transaction-Ordering Bias and MEV in Mysticeti DAG-based BFT Protocol.** I. Mirzaei, M. J. Amiri. [arXiv:2607.13378](#), 2026. ([Code](#))
- **Sample More, Reflect Less: Self-Refine and Reflexion Lose to Repeated Sampling at Equal Token Cost, from 1.5B to 7B.** I. Mirzaei. [arXiv:2607.28576](#), 2026.
- **Stateless Network-Aware Adaptive Bitrate Streaming over IPFS.** I. Mirzaei, S. J. Mojaveri, A. Najafizadeh. [arXiv:2606.29574](#), 2026.
- **Poster: Fishing in Deeper Waters: A Comprehensive MEV Analysis of DAG-Based Blockchains.** I. Mirzaei, M. J. Amiri. [North East Database Day 2026](#), UMass Boston.

PROFESSIONAL EXPERIENCE

Application Team Lead - Cross-Platform Developer, *AI Home Design* (Remote, BC, CA) Apr 2024 - Jul 2024

- Developed Android/iOS application from scratch using Flutter for AI-powered home design platform
- Led technical decisions for cross-platform architecture, reducing development time by 40% compared to native approach while serving global user base
- Implemented real-time rendering features achieving 60fps performance and seamless integration with AI design backend services

Software Engineer - Cross-Platform Developer, *Plates Company* (Remote, Paris, FR) Jul 2023 - Oct 2023

- Developed Android/iOS/PWA chat feature for peer-to-peer local food services platform
- Integrated 4 Firebase services (Auth, Firestore, Cloud Messaging, Storage) enabling real-time messaging with <100ms latency; zero critical bugs in production

Software Engineer - Cross-Platform Developer, *Vusic Company* Nov 2022 - Jul 2023

- Developed iOS and PWA applications from scratch for country's largest music streaming platform
- Performed performance analysis and optimization using Flutter DevTools, improving app load time by 35% and reducing memory usage by 25%
- Built various features for social media integration and music/video streaming, serving +200K total users and +21K daily active users

Flutter Developer Technical Lead, *Black Bit* Sep 2021 - Nov 2022

- Led team of 5 developers delivering 4 applications with 98% on-time delivery, reducing sprint time by 30% and improving team velocity by 40%
- Managed 10+ client projects; maintained CI/CD pipelines reducing deployment from 4 hours to 15 minutes using FastLane, shipping real-time communication and GPS tracking features across health-care, entertainment, and education apps

RESEARCH & TEACHING EXPERIENCE

- Research Assistant, *SeeMoRe Lab, Stony Brook University*

Jan 2025 - present

 - Researching MEV and transaction-ordering vulnerabilities in DAG-based BFT consensus protocols under [Prof. Mohammad Amiri](#); built a Rust benchmarking harness measuring ordering bias across Mysticeti, Bullshark, and related protocols
 - First-authored an arXiv preprint and an accepted poster (NEDB Day 2026) from this line of work, with further papers in progress ([Code](#))
 - **Mentored Simons Summer Research Fellow Heer Patel** at the PI's request: onboarded her onto the MEV project, broke the research into scoped weekly tasks, taught DAG-based BFT consensus and the experimental tooling, and guided her through to a published research poster; now co-authoring a follow-up paper with her
- Research Assistant, *Ethos Security & Privacy Lab, Stony Brook University*

Sep 2024 - Dec 2024

 - Conducting research on ML security, and systems under [Prof. Amir Rahmati](#)
 - Developing ML security and forensic analysis capabilities to trace model failures, vulnerabilities, and attacks back to their origin, including model versions, training data lineage, update history, etc.
- Teaching Assistant, *Stony Brook University & IUST*

Sep 2021 - present

 - **CSE 316 - Software Development (SBU)**: 120+ students/semester, evaluated 500+ assignments, 6 weekly office hours, 130+ technical interviews
 - **Software Engineering, Database Design, Programming (IUST)**: Mentored 50+ students, conducted code reviews, created tutorial content
- Research Assistant, *Data Science Lab, IUST*

Sep 2023 - Feb 2024

 - Developed cross-platform fraud detection SDK achieving 95% accuracy, analyzing 10K+ interaction patterns with 5 key behavioral indicators
 - Coached 2 students in Flutter/Dart cross-platform development, resulting in 2 successful project completions

SELECTED PROJECTS

- MacSai: Open-Source Mac Cleaner & Security Scanner

[GitHub](#) ★ 1.3K+

Free, Apple-notarized alternative to CleanMyMac built with Swift 6 and SwiftUI. Covers system cleanup, performance optimization, and malware scanning. Reached 1.3K+ GitHub stars with active open-source community adoption.
- MEV Arbitrage Bot on Sui Blockchain

Proprietary

Designed and built a personal MEV arbitrage bot targeting transaction ordering opportunities on Sui. Built with deep knowledge of Mysticeti consensus and Sui's transaction lifecycle. Source code confidential.
- Linear PBFT Banking System

[GitHub](#)

Fault-tolerant distributed banking system built on the Linear PBFT consensus algorithm, keeping transactions consistent across replicas through consensus-based state machine replication even under Byzantine (malicious) failures. **Earned #1 rank of the class based on performance and correctness** (SBU, 2025)
- Paxos Banking System

[GitHub](#)

A distributed scalable banking application implementing the Multi-Paxos consensus protocol for state machine replication, with comprehensive performance evaluation using the SmallBank benchmark. **Earned #1 of the class based on performance and correctness** (SBU, 2025)
- Telescope - Smart ABR Proxy for IPFS

[GitHub](#)

Built scalable adaptive bitrate proxy for IPFS with cache-aware quality adaptation. Architected stateless microservices using Go-Fiber, Prometheus metrics, and OpenTelemetry tracing. **Published as arXiv:2606.29574** (SBU, 2024)

SKILLS

Programming Languages	Python, Dart, C/C++, Rust, Go, Java, C#, JavaScript, SQL
Web & Mobile	Flutter, D3.js, HTML, FastAPI, RESTful APIs, Progressive Web Apps (PWA)
Frameworks & Tools	.NET, Echo, SDL, Docker, Rabbit-MQ, Git/GitHub, Firebase, Figma
Databases	SQL Server, Postgres, MySQL, MongoDB, Neo4j, Firestore
Systems & DevOps	Linux (Tracing, Debugging, Kernel Development), Nginx, CI/CD Pipelines, Version Control
Testing & Methodologies	Unit Testing, Integration Testing, Test-Driven Development, Agile/Scrum, Code Review
AI & Agentic Development	Claude, Claude Code, Agentic Coding, Agent Loop Engineering, AI Automation, Google Stitch

HONORS & AWARDS

- Best Project Awards - Linux Kernel Development & Mobile Networks courses, IUST, 2023
- Top 2% national University Entrance Exam (261/164,278), 2019 — Qualified for direct Master's admission (Top 10%), 2023
- Graduated from NODET (National Organization for Development of Exceptional Talents) with 4.0/4.0 GPA